

Ruchita Solanki

University of Calgary | <https://ruchita-0310.github.io/> | ruchita.solanki@ucalgary.ca

Profile

PhD researcher in environmental microbiology and microbial biotechnology with expertise in photosynthetic microorganisms, microbial consortia, genome-resolved metagenomics, metatranscriptomics, and functional genomics. Experienced in microbial cultivation, genome annotation, bioinformatics workflow development, and the study of extremophilic microorganisms in alkaline environments. Interested in translating microbial physiology and multi-omics research into sustainable biotechnology, clean energy, and biorefinery applications.

Education

Ph.D., Geosciences and Biological Sciences **Sep 2022 – Present**
University of Calgary, Calgary, Alberta, Canada

B.Sc., Biological Sciences **August 2016 – May 2020**
University of Cincinnati, Cincinnati, Ohio, USA

- 3.00/4.00 cumulative GPA.

High School Diploma, International Baccalaureate **March 2014 – Jan 2016**
The Galaxy School, Rajkot, Gujarat, India

- Diploma in Mathematics, English, Hindi, Biology, Psychology and Chemistry.

Research Experiences

Extraction of Volatile Lipids **Sept 2023 – Present**
University of Calgary – Calgary, AB

- Extracting fatty acids from cyanobacteria
- Analyzing and quantifying of lipids using GC-MS

Nanopore Sequencing of consortia **Aug 2023 – Present**
University of Calgary – Calgary, AB

- High molecular weight DNA extraction using phenol chloroform and CTAB
- Creating a bioinformatic pipeline for analyzing the data

Comparative Genomics Project **April 2023 – Aug 2023**
University of Calgary – Calgary, AB

- Bioinformatics project on “Ancestral Reconstruction of programmed cell death using comparative genomics approach”
- Writing python code to run bioinformatics algorithms to map a set of genes to existing annotated genome using cloud computing
- Using python to annotate genes present at each internal node

Characterization of a *Pseudomonas stutzeri* strain **Aug 2018 – May 2020**
Rowe Lab, University of Cincinnati – Cincinnati, OH

- Worked on electrochemically active microbes isolated from springs.
- 16S rRNA sequencing of bacteria.
- Measured the membrane potential using the concept of fluorescence microscopy as a proof of concept
- Performing genetic manipulation on bacteria via conjugation.
- Used bioinformatic software - Genious Prime, 16S Biodiversity Tool and FigTree

Academic Publications

- **Solanki, R.**, Wiszniak, E., Yi, L., Cabria, G., Strous, M., & Davila Aleman, F. D. (2026). Genome mining of alkaliphilic cyanobacterial consortia: identification of biosynthetic gene clusters in *Sodalinema* and associated heterotrophs. *Applied and Environmental Microbiology*, e00780-26. <https://doi.org/10.1128/aem.00780-26>
- Yi, L., **Solanki, R.**, Moll, M., Vadlamani, A., De la Hoz Siegler, H., & Strous, M. (2025). Toward sustainable phycocyanin production using halo-alkaliphilic cyanobacteria: from direct air capture of carbon dioxide to biorefinery. *Frontiers in Microbiology*, 16, 1618123. <https://doi.org/10.3389/fmicb.2025.1618123>
- Yi, L., **Solanki, R.**, & Strous, M. (2024). In search of the pH limit of growth in halo-alkaliphilic cyanobacteria. *Environmental Microbiology Reports*, 16(4), e13323. <https://doi.org/10.1111/1758-2229.13323>
- **Solanki, R.**, Bhatt, A., Patel, J., & Kothari, R. (2023). Evaluation of prebiotic utilization and probiotic properties of potential probiotic bacteria of food and human origin. *Food Science & Nutrition Technology*, 8(2), 1–12. <https://doi.org/10.23880/fsnt-16000297>
- **Solanki, R.**, & Rowe, A. (2019). Characterization of a *Pseudomonas stutzeri* strain from marine sediments capable of iron oxidation and electrode oxidation. *Undergraduate Scholarly Showcase*, 1(2).

Teaching Experiences

Teaching Assistant of Science Communication

Sept 2023 – Dec 2023

University of Calgary – Calgary, AB

- My role was multifaceted, involving not only assisting in lectures and grading assignments but also actively engaging with students to enhance their communication proficiency.

Teaching Assistant of the Biotechnology Lab

June 2021 – April 2022

Saurashtra University – Gujarat, India

- Aided first-year and second-year students of M.Sc. Biotechnology in the lab during practical sessions.

Assistant to Teaching Assistant of the Microbiology Lab

Jan 2020 – May 2020

University of Cincinnati – Cincinnati, OH

- Helped first-year and second-year students to master the skills of using the microscope, aseptic techniques, and other microbiological techniques in the lab.

Professional History

Microbiologist

Department of Biosciences, Saurashtra University, Rajkot

Feb 2021 – Aug 2022

- Isolation, screening, and identification of bacterial isolates from various types of natural samples such as soil, sewage, marine waters, and foods
- Isolation of plasmid DNA
- Aseptic transfer, preservation and maintenance of bacterial cultures
- Metagenomic and genomic DNA isolation
- Biotech techniques like RNAi, CRISPR-Cas9, cloning, PCR, and sequencing
- Various staining and streaking techniques

Bhatt Diagnostic Laboratories, Rajkot, India

June 2020 – Dec 2020

- Logged in the samples into the system.
- Looking at microbial growth and identifying the organism.
- Performing manual MIC
- Making a report based on the antibiotic tests.

Conferences and Symposiums

Canadian Society of Microbiologists (CSM) – Montreal, Canada **June 2025**

- Poster presentation “Cultivation and whole genome sequencing of cyanobacterial consortia enriched from Canadian alkaline soda lakes”

Lake Arrowhead Microbial Genomics – CA, USA **Sept 2024**

- Poster presentation “Growth and whole genome sequencing of diatom consortia enriched from Canadian alkaline soda lakes”

The Geological Society of America (GSA) – CA, USA **Sept 2024**

- Poster presentation “Growth and whole genome sequencing of cyanobacteria and diatom consortia enriched from Canadian alkaline soda lakes”

International E-Science Symposium at Christ College – Gujarat, IN **Jan 2022**

- Poster presentation “Potential Probiotic Bacteria with Human Fecal Origins”

Swedish South Asian Network for Fermented Foods (SASNET-FF) Conference **Sept 2021**

- Poster presentation “Isolation screening and identification of potential probiotic bacteria from human feces”

American Society for Microbiology – Ohio Branch, Cleveland, OH **Feb 2019**

- Poster presentation “Characterization of a *Pseudomonas stutzeri* strain from marine sediments capable of iron oxidation and electrode oxidation”

University of Cincinnati Scholarly Showcase **April 2019**

- Poster presentation “Growth Physiology of Electrode Oxidizing *Pseudomonas stutzeri* strain”

Honors & Awards

- Received funding of \$1600 from ISME as a travel grant **Aug 2026**
- Received funding of \$300 from Canadian Society of Microbiologists **June 2025**

- Received funding of \$11,000 from Alberta Graduate Excellence Scholarship **Sept 2024–2025**
- Received funding of US\$20,000 from Torrent Pharmaceuticals Ltd. **Sept 2022–Sept 2024**
- Dean's list, University of Cincinnati, Cincinnati. **April 2020**
- The Duke of Edinburgh's International Award Gold Standard, Rajkot. **April 2016**
- Counseling Skills and Major Psychotherapies, Rajkot. **Dec 2014**

Skills

- **Programming Languages:** Python, R, Markdown
- **Language:** English, Gujarati, Hindi, Spanish
- **GitHub repository:** <https://github.com/Ruchita-0310>